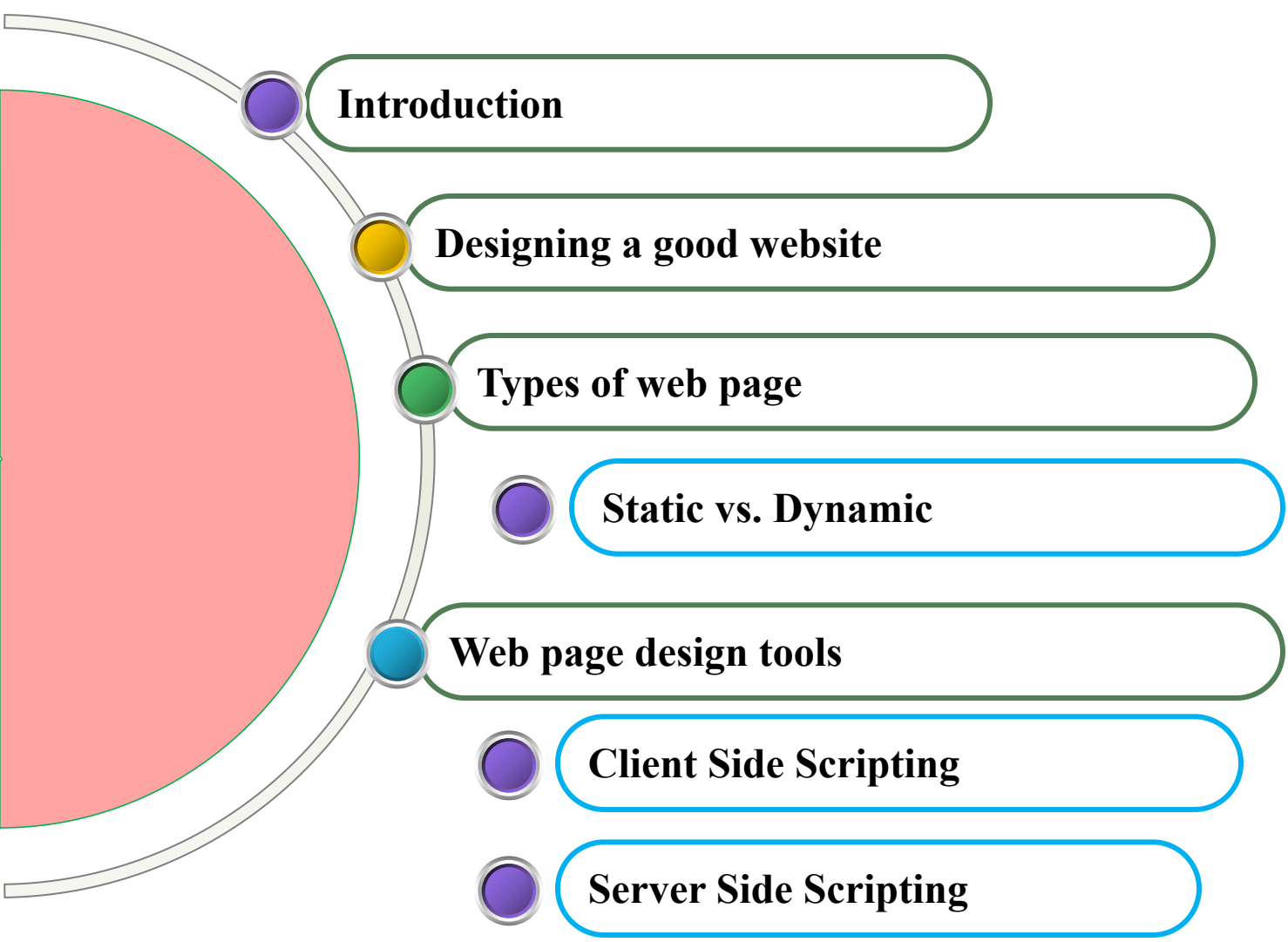


CHAPTER ONE

INTRODUCTION TO INTERNET PROGRAMMING

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Introduction

Internet

- Has been coined from two terms, **Interconnection** and **Network**
- Is a **huge network of computers** that links many different types of computers all over the world.
- It is a **network of networks** that share a common mechanism for addressing (identifying) computers for communications between two computers on the network
- A **Network** is simply a **group of computers that are connected together for sharing information and resources**.
 - Several such networks have been joined together across the globe to form what is called as the **Internet**.

There is no single, generally agreed-upon answer to the question because the Internet is different for each of us:

- It is a set of computers talking over fiber optics, phone lines, satellite links, and other media.
- It is a place where you can talk to your friends and family around the world.
- It is an ocean of resources waiting to be mined.
- It is a place to do research for your thesis or a business.
- It is a worldwide support group for any problem or need.
- It is a gold mine of professionals in all fields sharing information about their work.
- It is hundreds of libraries and archives that are open to your fingertips.

Who Owns the Internet

- No single entity owns the Internet.
- Any single person, service, corporation, university, or government does not fund it.
- Every person who makes a connection, every group who's Local Area Network (LAN) becomes connected, owns a slice of the Internet.
- The owner of the connected equipment therefore “owns” a piece of the Internet.
- The telephone companies “own” the pieces that carry the packets (blocks of information).
- The service providers “own” the packet routing equipment. So, while no one person or entity owns the Internet, all who use it or supply materials for it play a part in its existence.

Services of Internet

- May be four things, basically:
 1. Electronic Mail (E-Mail)
 2. World Wide Web (WWW)
 3. File Transfer Protocol (FTP)
 4. E-commerce

1. Electronic Mail [E-Mail]

- is a **paperless method of sending messages, notes or letters** from one person to another or even many people at the same time via the Internet.
- E-mail is very fast compared to the normal post.
- E-mail messages usually take only few seconds to arrive at their destination.
- You have the privilege of sending something extra even such as a file, graphics, images etc. along with your e-mail.
- **Advantage:** cheaper and faster

a) Features of E-Mail

- One-to-one or one-to-many communications
- Instant communications
- Physical presence of recipient is not required
- Most inexpensive mail service, 24-hours a day and seven days a week
- Encourages informal communication

b) Components of E-Mail Address

- E-mail is also **based upon the concept of a recipient address**.
- The email address provides all of the information required to get a message to the recipient from anywhere in the world.

Consider the e-mail ID **john@hotmail.com**

- In the example above, **"john" is the local part, which is the name of a mailbox** on the destination computer, where finally the mail will be delivered.
- **Hotmail** is the **mail server** where the mailbox "john" exists; **.com** is the type of organization on net, which is hosting the mail server.
- six main categories which are hosting the mail server:
 - **com** Commercial institutions or organization
 - **edu** Educational institutions
 - **gov** Government site
 - **mil** Military site
 - **net** Gateways and administrative hosts
 - **org** Private organizations

2. World Wide Web (WWW)

- It is also commonly known as ‘[The Web](#)’.
- The WWW is [hypertext based information retrieval tool](#).
- One can easily surf the Web by jumping from one document to another using the links in those documents.
- These documents can be in many formats, such as text, graphics, animation, sound and latest is video.
- All the information on Internet are presented to the user as a document or more popularly known as [Web Page](#).

- All these Web Pages are link to each other or even to section within a Web Page. And these links are known as **Hyper Links**.
- The group of web pages which are interlinked is called **web site**.
- The tool used to view these Web Pages on Internet is known as *Internet browser* or simply **browser**.
- **Hyperlink**, also called link – an element used to connect one Web page to another Web page that's located on the same server or different Web server located anywhere in the world

- www (world wide web)
 - A collection of websites
- A website
 - A collection of resources:
 - Web pages (static / dynamic)
 - Media files (images, animations, sound, ...)
 - Style files (CSS)
 - Documents (pdf, doc, txt, rtf, ...)
 - ...
 - Has a globally unique **name DNS** (Domain Name System)
 - Resolves a human friendly name (eg www.google.com) to a machine friendly IP address (eg 64.233.187.99)
 - “Phone book” of the Internet
 - E.g. www.sims.ddu.edu.et → 10.235.5.35
 - Stored on machines called **web servers**

3. FTP (File Transfer Protocol)

- an Internet utility software used to **upload and download files**.
- It gives access to directories or folders on remote computers and allows software, data and text files to be transferred between different kinds of computers.

The basic objectives of FTP are

- To give flexibility and promote sharing of computer programs, files and data
- To transfer data reliably and more efficiently over network
- To encourage implicit or indirect use of remote computers using Internet
- To shield a user from variations in file storage systems among hosts.

Internet Address (Web Address)

- Just like every house, every office, every location has an address;
- every page on the Internet has a unique address. This address is used to get the web page for user from Internet.
- Just as the address of a house or office is known as its postal address, the address on the Internet is known as URL (Uniform Resource Locator).

URL is like; <protocol>://<host>[:<port>][<path>][?<query>]

<http://www.nos.org/computers/internet/url.htm>

- The URL contains the components that specify the protocol, server, and pathname of an item.
- The **protocol** is followed by a colon (**http:**),
- the **server** is preceded by two slashes (**//www.nos.org**), and each segment of the **pathname** is preceded by a single slash (**/computers/internet/**).
- url.htm is a file name or webpage.
- A **protocol** is set of rules that tells the computer how to interpret the information at that address.

Designing a Good Website

What makes a good website

- Know your audience- what will they use your page for?
- Content- is the content that they want?
- Timely, current and relevant
- Accurate and trustworthy
- Appropriate links for your form
- Design for readability
- Design for attractiveness
- Design for easy of understanding
- Design for efficiency
- Design to meet your goals

Types of Web pages

- There are basically two main types of web pages: **Static** and **Dynamic**.

Static Web page

- Is one that is written in plain HTML and what is in the code of the page is what is displayed to the user.
- Each web page is a separate document and there are no databases or external files that are drawn upon.
- The only way to edit is to go into each page and edit HTML (do yourself using a web page editor or pay your web developer to updates)
- Useful for displaying data that doesn't change often

Advantages of Static Web page

- Flexibility is the main advantage of a static site
- Every page can be different if desired, to match the layout to different content, and the designer is free to put in any special
- Allows theming - for instance an author may want a different theme for a different book and associated pages or perhaps for a series of books, in order to match the cover designs or the context of the stories.
- Cost is generally lower up-front than a dynamic site.

Disadvantages of Static Web page

- The main problem is **to update the content**. Unless you are conversant with HTML and the design methods used in the site then you have to go back to the designer to have any content changes made.
- The second main problem is **scalability**. If you wish to sell products on your site and you have a lot of them then you may have to construct individual pages for each one, which can take considerable time, effort and cost.
- **Costs** - there are ongoing costs for updating the content.

Dynamic Web page

- Is one that is written using a server-side scripting language such as PHP, ASP, JSP, or Coldfusion.
- In a dynamic site the content is called in by the scripting language from other files or from a database depending on actions taken by the user.
- Can be controlled by a content management system
- Content Management System is a software application that can be used to manage the creation and modification of digital content.

Advantages of Dynamic Web page

- You can easily pull in information in an organized way
- You can also create a content management system - an interface which allows the client to input and manage data via a web-based series of administration pages.
- Reduce ongoing maintenance costs, b/c u will enable u to maintain your own content
- Make data management very efficient
- Much easier to update

Disadvantages of Dynamic Web page

- The design of a dynamic site is more fixed than a static one because many of the pages are essentially a template into which data and content is poured to create multiple pages of a similar type.
- A change to the page design will be evident on every page.
- Costs are higher initially than for a static site because they require more complex coding, as well as for maintenance
- Need to obtain web hosting which supports databases and dynamic languages

Static Vs Dynamic

Static Website	Dynamic Website
Prebuilt content is same every time the page is loaded.	Content is generated quickly and changes regularly.
It uses the HTML code for developing a website.	It uses the server side languages such as PHP, ASP.NET etc. for developing a website.
It sends exactly the same response for every request.	It may generate different HTML for each of the request.
The content is only changed when someone publishes and updates the file (sends it to the web server).	The page contains "server-side" code which allows the server to generate the unique content when the page is loaded.
Flexibility is the main advantage of static website.	Content Management System (CMS) is the main advantage of dynamic website.

Web page Design Issues

1. Users are scanners

- If you think a typical user will read the entire content of your web pages, you are wrong.
- No matter how much useful information you put into a webpage, a visitor will only spend a few seconds scanning it before they decide whether to leave or stay.
- If you want a visitor to read your text, be sure to make your point in the very first sentence of the page.
- After that you should try to keep them occupied with short paragraphs and interesting new headers all the way down the page.

2. Less is more

- Try to keep all sentences as short as possible.
- Try to keep your paragraphs as short as possible.
- Try to keep your pages as short as possible

3. Use a lot of space between your paragraphs

- Pages overload with text will kill your audience.

4. Don't place too much content on a single page

- If you have a lot to say, try to break your information into smaller chunks and place it on different pages.
- Don't expect any visitor to scroll all the way down to bottom of a page with thousands of words.

5. Navigation

- Try to create a navigation structure that is common for all the pages in your web.

6. Download speed

- A common mistake made by web designers is to develop a site on a local machine with direct access to the data, or to develop the site over a high speed internet connection.
- Sometimes developers are not aware of the fact that some of their pages take a long time to download.
- Before you publish any content **heavy pages**, make sure they are tested over a low speed modem connection.

7. Let your audience to speak

- Feedback from your users is very good thing.

Steps to Develop Website

- Once you have decided to establish a web site there are 3 steps to develop website.
 1. Obtain the appropriate **equipment**
 - Web Server – hardware and software
 2. Register Web Server to an **Internet Service Provider (ISP)**
 - Obtain the IP address and DNS address
 3. Develop the **contents**
 - Internet Programming

Website Designing Tools

- A. Client side scripting [HTML, CSS, JavaScript, Applet]
- B. Server side scripting [PHP, Asp.net, Ruby]
 - Notepad, Notepad++
 - Sublime
 - Dreamweaver
 - Fireworks
 - Microsoft FrontPage
 - Macromedia flash
 - Microsoft office publisher
 - WAMP or XAMP



END OF CHAPTER ONE

Next: Chapter: Two